Investigation of Nitrogen Enhanced NBTI Effect Using the Universal Prediction Model

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***Abstract*—The impact of nitrogen concentration on the negative bias temperature instability (NBTI) effect in p-MOSFETs with SiON gate dielectrics is investigated in this paper. The universal NBTI model considering the interface state generation and hole-trapping/detrapping mechanisms is used for modeling the nitrogen enhanced NBTI degradation. Increased nitrogen concentration increases the interface trap density and reduces the activation energy. Correspondingly, the critical model parameters such as the hole-capture time constant and the interface state generation rate are modified. The universality of NBTI model is verified by comparing with the measurement data under conditions of different nitrogen concentrations. Based on the universal NBTI model, the impact of nitrogen on interface-state generation and hole-trapping mechanisms are investigated respectively. The result shows that interface-state generation is more sensitive to nitrogen concentration variation than hole-trapping mechanism. Therefore, the nitrogen enhance NBTI degradation in the high frequency circuit is not as significant as that under low frequency or DC operation conditions.**

Keywords- nitrogen; NBTI; universal model; interface state generation; hole-trapping

I. INTRODUCTION

Nitrogen incorporated gate dielectrics is commonly used for increasing the dielectric constant and suppressing the direct-tunneling gate leakage [1-3]. However, nitrogen enhanced negative bias temperature instability (NBTI) effect becomes a considerable reliability issue for thin gate oxide, especially with high nitrogen concentration (N%) close to the oxide-substrate interface [4-6]. Therefore, in order to help circuit designers have a clear view and appropriate design evaluation of circuits based on the oxide nitride dielectrics processes, an NBTI model considering effects from different nitrogen concentrations is required for device reliability prediction and circuit aging simulation. In this paper, the universal NBTI model [7] is proposed to simulate the nitrogen enhanced NBTI effect by modifying corresponding model parameters depending on various N%. For this purpose, an automatic parameter extraction tool is developed to verify the universal model using the measured data. Based on the modeling analysis, the interface-state generation ($\Delta V_{\text{th_int}}$) is found much more sensitive to N% than the hole-trapping ($\Delta V_{\text{th_trap}}$). Moreover, as demonstrated in our previous work [8], $\Delta V_{\text{th_trap}}$ is considered as the domination mechanism in the short term region. Therefore, the high frequency AC NBTI effect is less impacted by the nitrogen concentration variation than the DC NBTI effect.

II. IMPACT OF NITROGEN CONCENTRATION ON NBTI MECHANISMS

Physics effects of nitrogen on NBTI are illustrated in Fig. 1. Higher concentration of nitrogen introduces more positive charged "K-center" defects at the oxide/Si interface [9]. Moreover, nitrogen incorporation is experimentally observed to reduce the activation energy for both interface-state generation and charge trapping [2, 3]. These two factors on the threshold voltage degradation were already considered in the NBTI model [7]. To describe the nitrogen enhanced NBTI effect, the model parameters which are closely associated to defect generation and hole-trapping/detrapping behaviors, such as existing interface trap density N_{trap0} , density of injected holes N_0, interface-state reaction rate R_{str}, reaction time constant τ_{str} and hole-capture time constant τ_{c}, are expected to be functions of N%.

For clarity, core equations of the universal NBTI model are listed as Eqs. (1) - (5), and parameter definitions are summarized Table. I.

Stress:

$$\Delta V_{th} = \Delta V_{th_\text{int}} + \Delta V_{th_trap} + \Delta V_{th_ox} \tag{1}$$

Interface state generation

$$\Delta V_{th_\text{int}}(t) = \left(q / C_{\text{ox}}\right) \cdot R_{str} \cdot \log(1 + t / \tau_{str}) \tag{2}$$

Hole trapping at the interface

$$\Delta V_{\text{th_trap}}(t) = \left(qN_{\text{trap0}} / C_{\text{ox}}\right) \cdot \left[1 - \exp\left(-t / \tau_{\text{c}}\right)\right] \tag{3}$$

Oxide traps

$$\Delta V_{\text{th_ox}}(t) = \left(qN_0 / C_{\text{ox}}\right) \cdot t^{1/6} \tag{4}$$

Recovery:

$$\begin{aligned}\Delta V_{\text{th}} &= \Delta V_{\text{th_trap}}{}^{\max} \exp\left(-t / \tau_{\text{e}}\right) \\ &\quad + (\Delta V_{\text{th_int}}{}^{\max} + \Delta V_{\text{th_ox}}{}^{\max})(1 + t / \tau_{rec})^{-m}\end{aligned} \tag{5}$$

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Table I. The universal NBTI model and parameters.

Parameter	Meaning
τ_{str} [s]	Interface state generation time constant
τ_{rec} [s]	Interface state recovery time constant
τ_e [s]	Hole emission time constant
N_0 [cm^{-3}]	Density of holes injected into oxide
N_{trap0} [cm^{-2}]	Interface trap density
R_{str} [cm^{-2}]	Interface state reaction rate coefficient
τ_c [s]	Hole capture time constant
m	Recovery time exponent

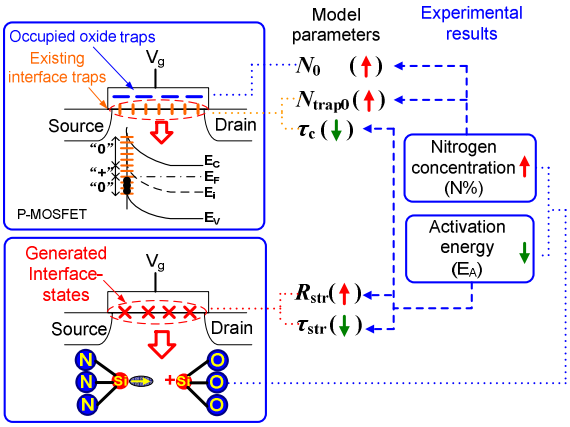

Fig. 1. Schematic view of nitrogen enhanced NBTI effect. The increased nitrogen density (N%) is expected to increase the oxide trap density (N_0) and interface trap density (N_{trap0}). Meanwhile, incorporation of nitrogen reduces the activation energy of hole-trapping and interface-state generation, which indicates not only the decreased hole-capture time constant (τ_c) and reaction time constant (τ_{str}), but also increased interface-state generation rate (R_{rec}).

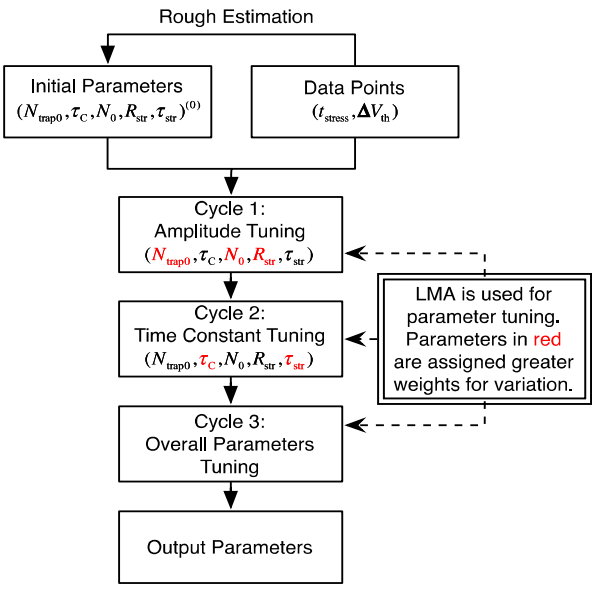

Fig. 2. Flow chart of the automatic parameter extraction (APE) tool. (N_0, N_{trap0}, τ_c, τ_{str}, R_{rec})$^{(0)}$ indicates the initial parameter values before stress process.

III. Model Parameter Extraction

In order to improve calculation efficiency, an automatic parameter extraction (APE) tool is developed. A modified Levenberg-Marquardt Algorithm (LMA) is leveraged for parameter extraction. The parameters are categorized into two types: Type I, parameters related to amplitude of ΔV_{th}, and Type II, the time constants. For NBTI stress, N_{trap0}, N_0 and R_{str} are Type I parameters, while τ_{str} and τ_c are Type II parameters. As illustrated in Fig. 2, the parameter extraction process is divided into three cycles. In Cycle 1, Type I parameters are given higher weights. In Cycle 2, Type II parameters are given higher weights. In Cycle 3, the parameters are assigned equal weights and overall parameter tuning is made. Such APE tool not only greatly improves the efficiency of data fitting but also improves accuracy. Comparisons between calculation and measured data [10, 11] are shown in Fig. 3.

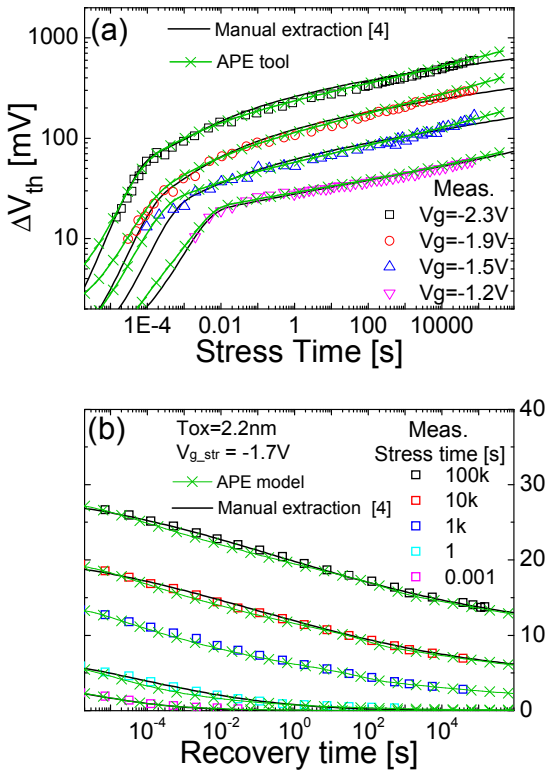

Fig. 3. Comparison between APE tool and manual extraction under (a) stress conditions with various biases [10] and (b) recovery conditions after various stress time [11].

IV. Modeling of Nitrogen Enhanced NBTI Effect

With the APE tool, the model parameters are accurately extracted from measured data with N= 20%, 35% and 42% [2], as shown in Fig. 4 The extracted model parameters are listed in Table. II. Increased N% introduces larger interface-trap density N_{trap0} [12], but does not evidently impact the oxide trap density N_0, which is relative to the trap assistant gate leakage [13, 14]. Such phenomenon has been demonstrated experimentally with the plasma annealing treatment [1]. Decreased time constants (τ_c and τ_{str}) and increased reaction rate (R_{rec}) are mainly due to the reduced activation energy in SiON dielectric [15]. The measurement results confirm the expectation illustrated in Fig. 1, and further verified the universal NBTI model under various processes.

Temperature is another significant impact issue to enhance the NBTI effect. As shown in Fig. 5, the nitrogen concentration dependence of model parameters is compared at different temperatures of 55°C and 125°C. Similar to the

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gate leakage, the existed interface-trap density (N_{trap}) is insensitive to the temperature shift Fig. 5(a). Moreover, increased temperature provides higher energy to enhance the vibrational property of chemical bonds located at the interface [16, 17]. Therefore, at higher temperature, the silicon bonds are easier to be broken and holes are easier to be captured by the defects, indicating smaller time constant and larger interface-state reaction rate and smaller time constant. Such hypothesis is confirmed by Fig. 5(b)-(d).

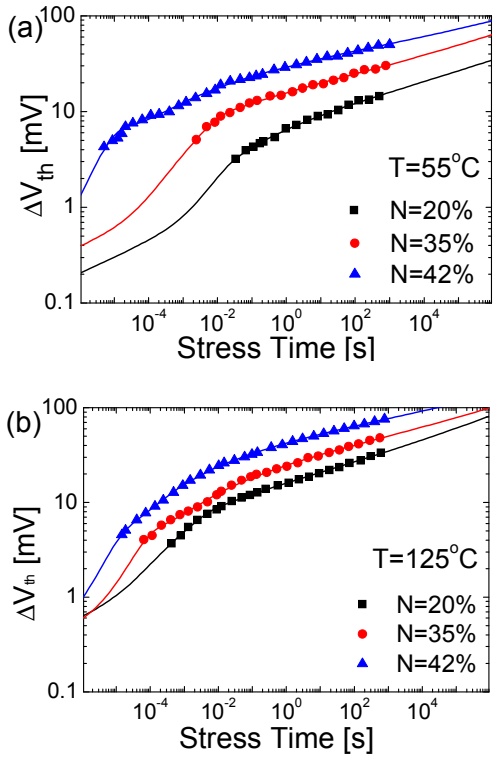

Fig.4. Nitrogen enhanced NBTI effect reproduced by APE tool with N%=20%, 35% and 42% at (a) 55°C and (b) 125°C. Data is from [2]. The model parameters are extracted using the automatic parameter extraction tool (APE).

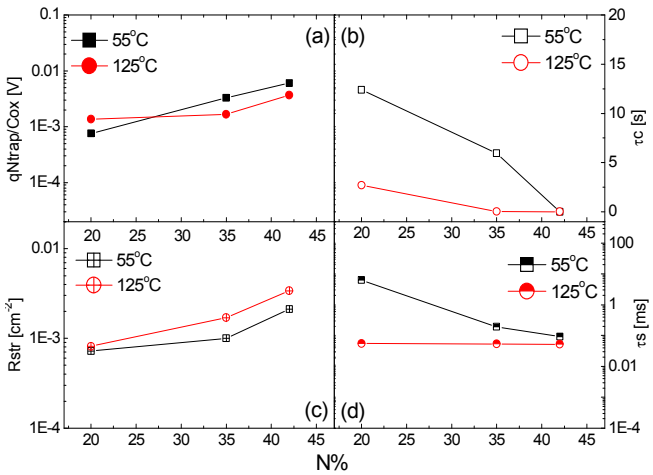

Fig. 5. Nitrogen concentration dependence of model parameters at temperatures of 55°C and 125°C, including (a) interface trap density N_{trap}., (b) hole-trapping time constant τ_c, (c) interface-state generation rate R_{str}, and (d) interface state generation time constant τ_s.

Table II. Nitrogen dependence of model parameters.

T/°C	N%	N_{trap0}/cm^{-2}	τ_c/ms	N_0/cm^{-3}	R_{str}/cm^{-2}	τ_s/ms
55	20	0.76e-3	12.4	2.04e-3	0.723e-3	6.403
	35	3.30e-3	5.93	3.84e-3	0.999e-3	0.193
	42	6.04e-3	0.0061	3.37e-3	2.120e-3	0.092
125	20	1.37e-3	2.69	6.13e-3	0.815e-3	0.056
	35	1.68e-3	0.019	6.44e-3	1.699e-3	0.054
	42	3.69e-3	0.0092	5.23e-3	3.386e-3	0.053

The individual contributions of the interface-state generation (ΔV_{th_int}), the hole-trapping in the existing traps (ΔV_{th_trap}) and oxide traps (ΔV_{th_ox}) to the threshold voltage shift (ΔV_{th}) are plotted in Fig. 6(a) based on the parameters extracted from Fig. 4(a). Nitrogen dependences of the three mechanisms are investigated in Fig. 6(b). Enhancement of ΔV_{th_int}, ΔV_{th_trap} and ΔV_{th_ox} with N=35% and 42% are calculated by subtracting the value with N=20%. ΔV_{th_int} shows the greatest sensitivity to nitrogen incorporation owning to the evident shift of R_{rec} and τ_{str}, especially in the long term region. However, less impact of nitrogen is observed on ΔV_{th_trap}. The nitrogen enhanced ΔV_{th} with N=35% and 42% are plotted in Fig. 7(a) Except for one observation that higher N% results in larger ΔV_{th}, the N% dependence in the short term region is much weaker than in the long term region. This is because that ΔV_{th_trap} and ΔV_{th_ox}, which feature less sensitivity to N%, are much more responsible than ΔV_{th_int} for the short term ΔV_{th} (Fig. 7 (b)) [8]. Therefore, high frequency AC NBTI effect should be less dependent on nitrogen distribution than DC NBTI effect. Such expectation has been strongly supported by the AC NBTI experiments [3].

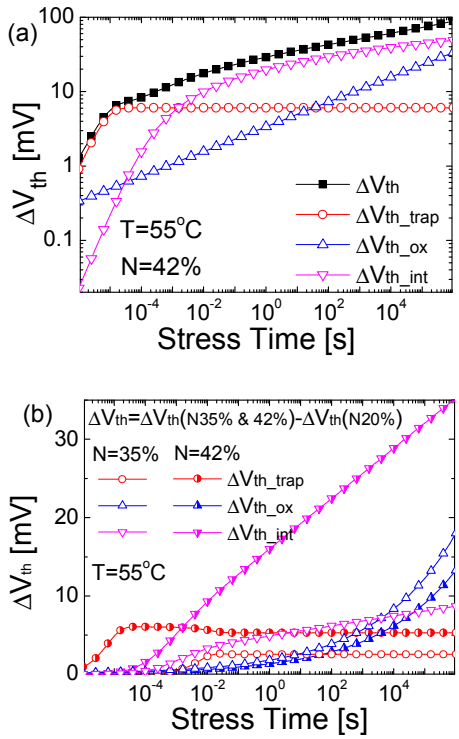

Fig. 6. (a) Individual contribution of the hole-trapping (ΔV_{th_trap}), the oxide traps (ΔV_{th_ox}) and the interface-state generation (ΔV_{th_int}) to the threshold voltage degradation (ΔV_{th}). (b) Nitrogen enhanced ΔV_{th_trap}, ΔV_{th_ox} and ΔV_{th_int} with N=35% and 42% compared to the value with N=20%. Greatest impact of N% is observed on ΔV_{th_int}.

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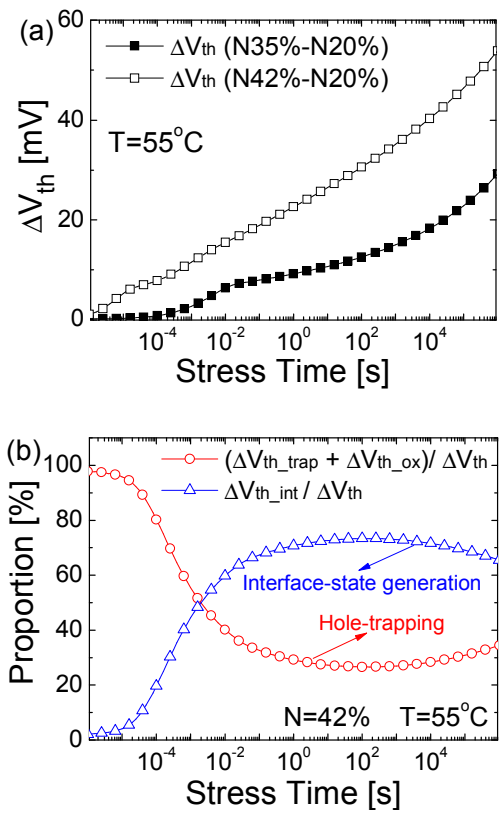

Fig. 7. (a) ΔV_{th} of N=35% and 42% compared to that of N=20%. The nitrogen enhanced NBTI degradation is more serious in the long term region than in the short term region, indicating that nitrogen incorporation results in larger degradation under long term DC condition than AC condition. (b) Contributions of interface-state generation and hole-trapping to the NBTI degradation. The hole-trapping dominates the short term region but less impacted by the nitrogen incorporation.

V. Conclusions

The nitrogen enhanced NBTI effect is proposed to be described by the universal NBTI model. N% dependence of the model parameters is investigated with the aid of a newly developed automatic parameter extraction tool. Moreover, the impact of nitrogen on the hole-trapping is much less than on the interface-state generation. Therefore, AC NBTI effect shows less sensitivity to N% than DC NBTI effect since the hole-trapping is dominant in the short term region.

Acknowledgement

The work as supported by the Hong Kong's University Grant Committee via the Area of Excellence project AoE-P04-08 and by the Shenzhen Science and Technology Project(JCYJ20140509093817685)

References

[1] K. Sakuma, D. Matsushita, K. Muraoka and Y. Mitani, "Investigation of Nitrogen-Originated NBTI Mechanism in SiON with High-Nitrogen Concentration,", IEEE International Reliability Physics Symposium (IRPS), pp. 454-460, 2006.

[2] S. Mahapatra, V. D. Maheta, A. E. Islam, and M. A. Alam, "Isolation of NBTI Stress Generated Interface Trap and Hole-Trapping Components in PNO p-MOSFETs," IEEE Transactions on Electron Devices, vol. 56, no. 2, pp. 236-242, 2009.

[3] Y. Mitani, H. Satake, and A. Toriumi, "Influence of Nitrogen on Negative Bias Temperature Instability in Ultrathin SiON," IEEE Transactions on Device and Materials Reliability, vol. 8, no. 1, pp. 6-13, 2008.

[4] S. S. Tan, T. P. Chen, C. H. Ang, and L. Chan, "Atomic Modeling of Nitrogen Neighboring Effect on Negative Bias Temperature Instability of pMOSFETs," IEEE Electron Device Letters, vol. 25, no. 7, pp. 504-506, 2004

[5] Shyue Seng Tan, T. P. Chen, C. H. Ang, and L. Chan, "Relationship between interfacial nitrogen concentration and activation energies of fixed-charge trapping and interface state generation under bias-temperature stress condition," Applied Physics Letters, vol. 82, pp. 269-271, 2003.

[6] S. Deora, V. D. Maheta, A. E. Islam, M. A. Alam, and S. Mahapatra, "A Common Framework of NBTI Generation and Recovery in Plasma-Nitrided SiON p-MOSFETs," IEEE Electron Device Letters, vol. 30, no. 9, pp. 978-980, 2009

[7] C. Ma, H. J. Mattausch, K. Matsuzawa, S. Yamaguchi, T. Hoshida, M. Imade, R. Koh, T. Arakawa, and M. Miura-Mattausch, "Universal NBTI Compact Model for Circuit Aging Simulation under Any Stress Conditions," IEEE Transactions on Device and Materials Reliability, vol. 14, no. 3, pp. 818-825, 2014.

[8] C. Ma, H. J. Mattausch, M. Miyake, T. Iizuka, K. Matsuzawa, S. Yamaguchi, T. Hoshida, M. Imade, R. Koh, T. Arakawa, J. He, and M. Miura-Mattausch, "Compact Reliability Model for Degradation of Advanced p-MOSFETs Due to NBTI and Hot-Carrier Effects in the Circuit Simulation," IEEE International Reliability Physics Symposium (IRPS), pp. 2A.3.1-2A.3.6, 2013.

[9] Patrick M. Lenahan, Jason P. Campbell, Anand T. Krishnan, and Srikanth Krishnan, "A Model for NBTI in Nitrided Oxide MOSFETs Which Does Not Involve Hydrogen or Diffusion," IEEE Transactions on Device and Materials Reliability, vol. 11, no. 2, pp. 219-266, 2011.

[10] Z. Ji, L. Lin, J. F. Zhang, B. Kaczer, and G. Groeseneken, "NBTI Lifetime Prediction and Kinetics at Operation Bias Based on Ultrafast Pulse Measurement," IEEE Transactions on Electron Devices, vol. 57, no. 1, pp. 228-237, 2010.

[11] H. Reisinger, O. Blank, W. Heinrigs, A. Mühlhoff, W. Gustin, and C. Schlünder, "Analysis of NBTI Degradation and Recovery-Behavior Based on Ultra Fast Vt-Measurements," IEEE International Reliability Physics Symposium, (IRPS), pp. 448-453, 2006.

[12] G. Gupta, S. Mahapatra, L. Leela Madhav, D. Varghese, K. Ahmed and F. Nouri, "Interface-Trap Driven NBTI for Ultrathin (EOT~12Å) Plasma and Thermal Nitrided Oxynitrides," IEEE International Reliability Physics Symposium (IRPS), pp. 731-732, 2006.

[13] J. T. Ryan, P. M. Lenahan, A. T. Krishnan, and S. Krishnan, "Energy resolved spin dependent tunneling in 1.2 nm dielectrics," Applied Physics Letters, vol. 95, no. 103503, 2009.

[14] Jonghwan Lee, Gijs Bosman, Keith R. Green, and D. Ladwig, "Model and Analysis of Gate Leakage Current in Ultrathin Nitrided Oxide MOSFETs," IEEE Transactions on Electron Devices, vol. 49, no. 7, pp. 1232-1241, 2002

[15] S. Mahapatra, A. E. Islam, S. Deora, V. D. Maheta, K. Joshi, A. Jain2 and M. A. Alam, "A Critical Re-evaluation of the Usefulness of R-D Framework in Predicting NBTI Stress and Recovery," IEEE International Reliability Physics Symposium (IRPS), pp. 614-623, 2011

[16] Gregor Pobegen and Tibor Grasser, "On the Distribution of NBTI Time Constants on a Long, Temperature-Accelerated Time Scale," IEEE Transactions on Electron Devices, vol. 60, no. 7, pp. 2148-2155, 2013

[17] Alain Bravaix, Y. Mamy Randriamihaja, V. Huard, D. Angot, X. Federspiel, W. Arfaoui, P. Mora, F. Cacho, M. Saliva, C. Besset, S. Renard, D. Roy, E. Vincent, "Impact of the gate-stack change from 40nm node SiON to 28nm High-K Metal Gate on the Hot-Carrier and Bias Temperature damage," IEEE International Reliability Physics Symposium (IRPS), pp. 2D.6.1-2D.6-9, 2013.

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